Data Science

Ensemble Models: Bagging & Random Forests

Advanced Modeling



GENERAL ASSEMBLY

Lesson roadmap

Welcome + Introduction

Ensemble Methods

Bootstrapping

Out-of-bag-error and feature importance

Random Forests

Bring It Home

Learning objectives



1. Define ensemble models and when to use them.
2. Describe how and why decision trees can be improved using bagging and random forests.
3. Fit and evaluate bagged decision trees and random forests.
4. Understand how to calculate out-of-bag error.
5. Explain how to extract the most important predictors in a random forest model.



Warm Up

Same Data, Different Trees

© 2025 General Assembly

DISCUSSIONS: Avoiding overfitting

Avoiding overfitting

- You're training a single decision tree and it overfits badly.
- What are some intuitive ways you might reduce overfitting **without** touching the model architecture?

 © 2025 General Assembly

Ensemble Methods

A problem...

The (white box) models we've seen so far may not be good enough for what we are trying to accomplish.

So what can we do?

We could use **neural networks**, but they require **a lot** of labeled training data to be effective.

One thing we can do to improve performance is to use **ensemble learning**.

 © 2025 General Assembly

Ensemble models

Instead of trying to get one very accurate model, **we can combine the predictions of many weaker models** to make one **ensemble model**.

 © 2025 General Assembly

How much does Penelope weigh?

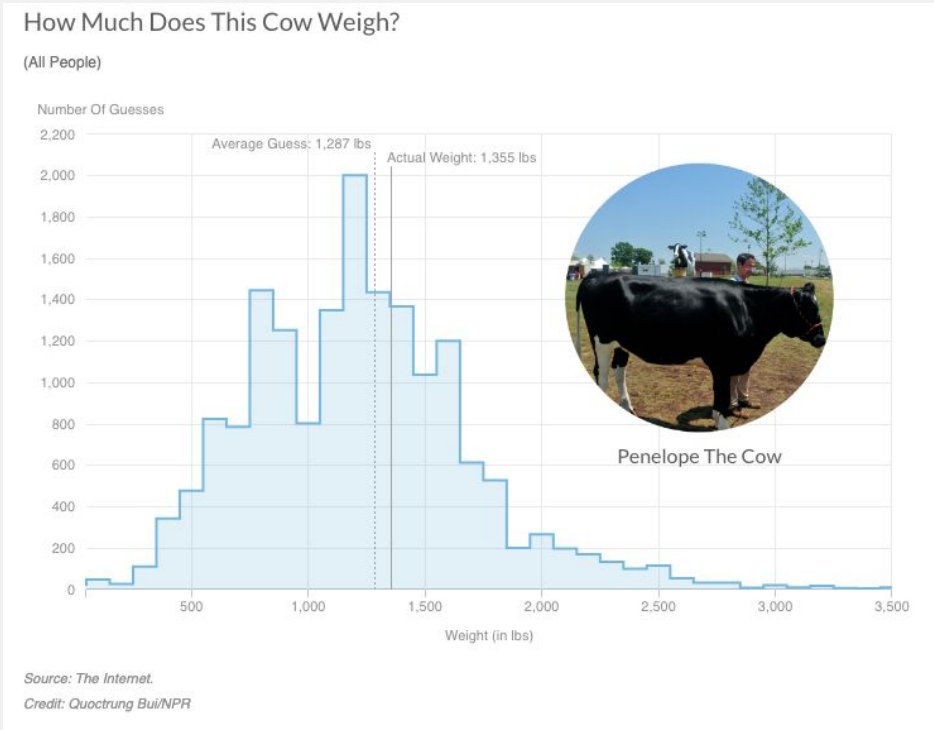


 © 2025 General Assembly

Polling a crowd







 © 2025 General Assembly

The Wisdom of the Crowds

Individuals are dumb, but crowds are smart!

It's an actual statistical fact that, instead of relying on one "ok" model's opinion, you'll get better results if you aggregate many "ok" models together!

This act of **ensembling** reduces the overall variance of your model. Since decision trees tend to overfit, they're a prime opportunity for ensembling.

While less effective, aggregating "dumb" learners (i.e., underfit models) also tends to improve model fit! Win-win!

 © 2025 General Assembly

How can we create ensemble models?

There are **3 main ensemble learning methods**:

1. Bagging
2. Boosting
3. Stacking

 © 2025 General Assembly

Bagging

Ensemble Methods

Bagging

We can simulate the wisdom of the crowds to improve our “simple” decision trees. The method of **b**ootstrap **agg**regat**ing**, or **bagging**, works as follows:

1. Take a sample of your data
2. Fit a model ***M*** (in our case a decision tree) to that data
3. Repeat this process ***B*** times (often $B = 500$, but it’s arbitrary)

You now have a set of models $M_1, ..., M_B$.

How can we use it?

 © 2025 General Assembly

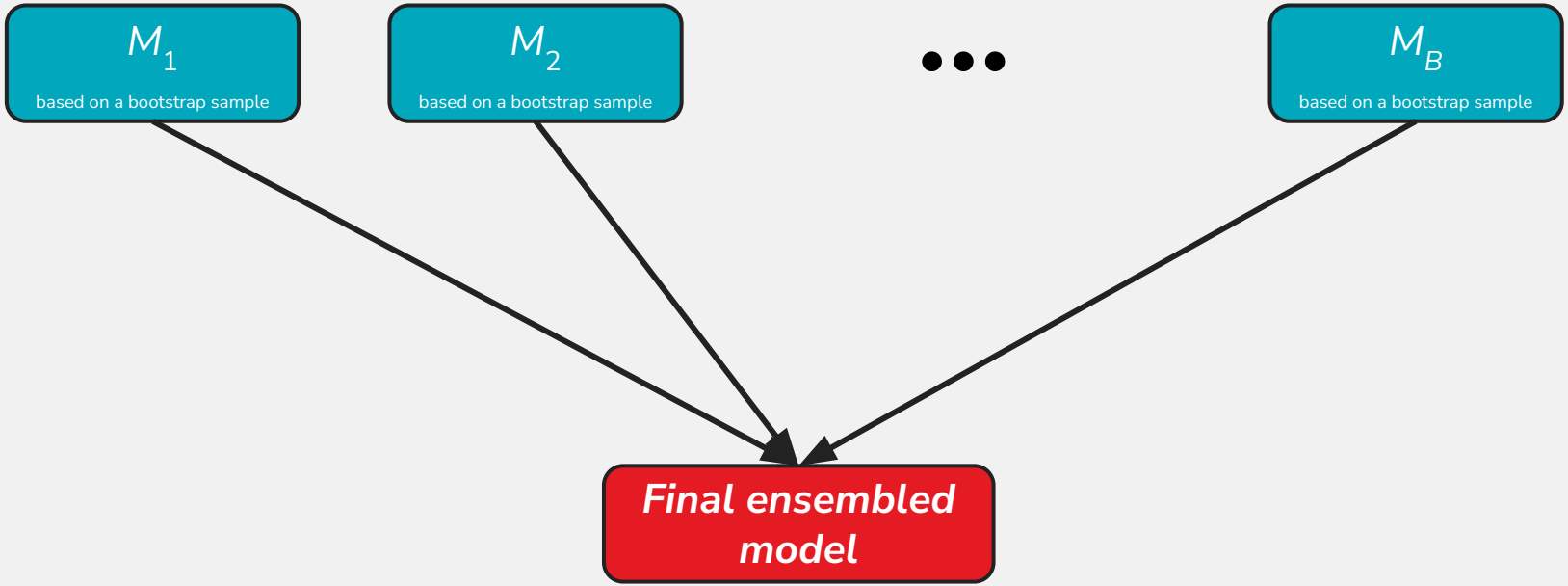
Bagging Prediction

For a new sample, compute its predictions for each model you have:

- For **regression** tasks, the final prediction is the **mean** of these predictions.
- For **classification** tasks, the final prediction is the **majority vote** of these predictions.

 © 2025 General Assembly

Bagging in a nutshell



 © 2025 General Assembly

How do we take samples of data?

We can simulate the wisdom of the crowds to improve our “simple” decision trees. The method of **b**ootstrap **agg**regat**ing**, or **bagging**, works as follows:

1. Create a bootstrap sample of your data
2. Fit a model ***M*** (in our case a decision tree)
3. Repeat steps 1-2 ***B*** times (often $B = 500$, but it’s arbitrary)

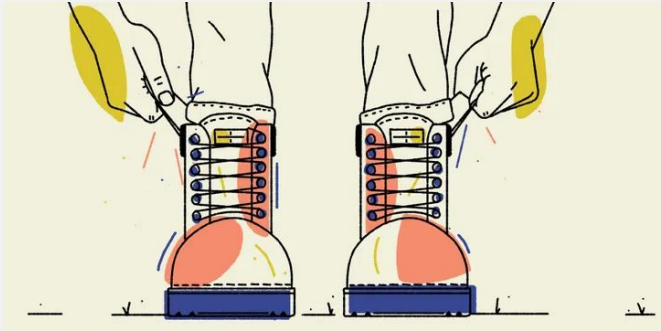
You now have a set of models $M_1, ..., M_B$.

 © 2025 General Assembly

How do we take samples of data?

We use **Bootstrapping**.

A **bootstrap sample** is a **random sample with replacement**. It has the same size as the original sample but might duplicate some of the original observations.



*Sidepoint: the **Bootstrap Technique** is a powerful resampling method used in statistics to estimate the sampling distribution of a statistic (like a mean, median, regression coefficient, etc.) by repeatedly sampling with replacement from the observed data.*

Source

 © 2025 General Assembly

For example...

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} & x_{15} & x_{16} \\ x_{21} & x_{22} & x_{23} & x_{24} & x_{25} & x_{26} \\ x_{31} & x_{32} & x_{33} & x_{34} & x_{35} & x_{36} \\ x_{41} & x_{42} & x_{43} & x_{44} & x_{45} & x_{46} \\ x_{51} & x_{52} & x_{53} & x_{54} & x_{55} & x_{56} \\ x_{61} & x_{62} & x_{63} & x_{64} & x_{65} & x_{66} \\ x_{71} & x_{72} & x_{73} & x_{74} & x_{75} & x_{76} \\ x_{81} & x_{82} & x_{83} & x_{84} & x_{85} & x_{86} \\ x_{91} & x_{92} & x_{93} & x_{94} & x_{95} & x_{96} \end{bmatrix}$$

Our data set has:

- 9 observations
- 6 features

 © 2025 General Assembly

Bootstrap Simulation... Go!

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} & x_{15} & x_{16} \\ x_{21} & x_{22} & x_{23} & x_{24} & x_{25} & x_{26} \\ x_{31} & x_{32} & x_{33} & x_{34} & x_{35} & x_{36} \\ x_{41} & x_{42} & x_{43} & x_{44} & x_{45} & x_{46} \\ x_{51} & x_{52} & x_{53} & x_{54} & x_{55} & x_{56} \\ x_{61} & x_{62} & x_{63} & x_{64} & x_{65} & x_{66} \\ x_{71} & x_{72} & x_{73} & x_{74} & x_{75} & x_{76} \\ x_{81} & x_{82} & x_{83} & x_{84} & x_{85} & x_{86} \\ x_{91} & x_{92} & x_{93} & x_{94} & x_{95} & x_{96} \end{bmatrix}$$

x2

x2

x2

x2

Our bootstrap sample has:

- 9 observations
- 6 features

 © 2025 General Assembly

Bootstrap Simulation... Go!

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} & x_{15} & x_{16} \\ x_{21} & x_{22} & x_{23} & x_{24} & x_{25} & x_{26} \\ x_{31} & x_{32} & x_{33} & x_{34} & x_{35} & x_{36} \\ x_{41} & x_{42} & x_{43} & x_{44} & x_{45} & x_{46} \\ x_{51} & x_{52} & x_{53} & x_{54} & x_{55} & x_{56} \\ x_{61} & x_{62} & x_{63} & x_{64} & x_{65} & x_{66} \\ x_{71} & x_{72} & x_{73} & x_{74} & x_{75} & x_{76} \\ x_{81} & x_{82} & x_{83} & x_{84} & x_{85} & x_{86} \\ x_{91} & x_{92} & x_{93} & x_{94} & x_{95} & x_{96} \end{bmatrix}$$

x3

x2

Our bootstrap sample has:

- 9 observations
- 6 features

 © 2025 General Assembly

Bootstrap Simulation... Go!

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} & x_{15} & x_{16} \\ x_{21} & x_{22} & x_{23} & x_{24} & x_{25} & x_{26} \\ x_{31} & x_{32} & x_{33} & x_{34} & x_{35} & x_{36} \\ x_{41} & x_{42} & x_{43} & x_{44} & x_{45} & x_{46} \\ x_{51} & x_{52} & x_{53} & x_{54} & x_{55} & x_{56} \\ x_{61} & x_{62} & x_{63} & x_{64} & x_{65} & x_{66} \\ x_{71} & x_{72} & x_{73} & x_{74} & x_{75} & x_{76} \\ x_{81} & x_{82} & x_{83} & x_{84} & x_{85} & x_{86} \\ x_{91} & x_{92} & x_{93} & x_{94} & x_{95} & x_{96} \end{bmatrix}$$

x3

x2

Our bootstrap sample has:

- 9 observations
- 6 features

 © 2025 General Assembly

Bootstrap Simulation... Go!

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} & x_{15} & x_{16} \\ x_{21} & x_{22} & x_{23} & x_{24} & x_{25} & x_{26} \\ x_{31} & x_{32} & x_{33} & x_{34} & x_{35} & x_{36} \\ x_{41} & x_{42} & x_{43} & x_{44} & x_{45} & x_{46} \\ x_{51} & x_{52} & x_{53} & x_{54} & x_{55} & x_{56} \\ x_{61} & x_{62} & x_{63} & x_{64} & x_{65} & x_{66} \\ x_{71} & x_{72} & x_{73} & x_{74} & x_{75} & x_{76} \\ x_{81} & x_{82} & x_{83} & x_{84} & x_{85} & x_{86} \\ x_{91} & x_{92} & x_{93} & x_{94} & x_{95} & x_{96} \end{bmatrix}$$

x2

x2

x2

Our bootstrap sample has:

- 9 observations
- 6 features

 © 2025 General Assembly

Bootstrap Simulation... Go!

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} & x_{15} & x_{16} \\ x_{21} & x_{22} & x_{23} & x_{24} & x_{25} & x_{26} \\ x_{31} & x_{32} & x_{33} & x_{34} & x_{35} & x_{36} \\ x_{41} & x_{42} & x_{43} & x_{44} & x_{45} & x_{46} \\ x_{51} & x_{52} & x_{53} & x_{54} & x_{55} & x_{56} \\ x_{61} & x_{62} & x_{63} & x_{64} & x_{65} & x_{66} \\ x_{71} & x_{72} & x_{73} & x_{74} & x_{75} & x_{76} \\ x_{81} & x_{82} & x_{83} & x_{84} & x_{85} & x_{86} \\ x_{91} & x_{92} & x_{93} & x_{94} & x_{95} & x_{96} \end{bmatrix}$$

x2

x2

Our bootstrap sample has:

- 9 observations
- 6 features

 © 2025 General Assembly

Bagging

Pros:

- Ensembles trees to reduce their extremely high variance
- Averaging unbiased models results in an unbiased model... with lower variance!

Cons:

- Since the bootstrapped models are based on largely the same data, the resulting models will be *correlated!*
- Averaging correlated variables limits how much we reduce the variance by!

 © 2025 General Assembly

Bagging

Let's see it for ourselves!

- ☐ Open **02-ensembles-bagging-random-forest.ipynb**



 © 2025 General Assembly

Evaluating models

Out-of-sample-error

Estimating Out-of-Sample Error

For bagged models, out-of-sample error can be estimated without using **train/test split** or **cross-validation**!

For each tree, the **unused observations** are called "out-of-bag" observations.

 © 2025 General Assembly

Calculating "out-of-bag error"

1. For each observation in the training data, predict its response value using **only** the trees in which that observation was out-of-bag.
2. Average those predictions (for regression) or take a vote (for classification).
3. Compare all predictions to the actual response values in order to compute the out-of-bag error.

When **B** is sufficiently large, the **out-of-bag error** is an accurate estimate of **out-of-sample error**.

 © 2025 General Assembly

Estimating Feature Importance

Bagging increases **predictive accuracy** but decreases **model interpretability** because it's no longer possible to visualize the tree to understand the importance of each feature. This is sometimes called a **black box model**.

But we can still obtain an overall summary of **feature importance** from bagged models:

1. **Bagged regression trees**: Calculate the total amount that **MSE** decreases due to splits over a given feature, averaged over all trees
2. **Bagged classification trees:** Calculate the total amount that **Gini index** decreases due to splits over a given feature, averaged over all trees

 © 2025 General Assembly

Estimating Out-of-Sample Error

Let's see it for ourselves!

- ☐ Open **02-ensembles-bagging-random-forest.ipynb**



 © 2025 General Assembly

Random Forests

A specific bagging method

Bootstrapping

We just introduced bagging: **b**ootstrap **agg**regat**ing**. This involved:

1. Getting a bootstrap sample
2. Fitting a model
3. Repeating 1 and 2 many times (B = 100 or 500 times)
4. The final model is the average of all B models.

 © 2025 General Assembly

The Random Forest Model

Random forest seeks to improve the variance reduction of bagging by decorrelating the tree without adding too much variance back.



 © 2025 General Assembly

The Random Forest Model

Random forest seeks to improve the variance reduction of bagging by decorrelating the tree without adding too much variance back.

We do this by adding another source of randomness:

Before each split, select a random subset of the input variables as candidates for splitting.

(Citation: ESL)



 © 2025 General Assembly

How many variables to select at each split?

The original authors of the random forest (Breiman et al, 2001) recommend:

- $\sqrt{p}$ for classification with minimum node size of 1 (where p is the number of features)
- $p/3$ for regression with minimum node size of 5

And these recommendations are generally good. The authors of ESL, on the other hand, **recommend you treat them as tuning parameters** as the best value will depend on the problem.

 © 2025 General Assembly

What's the point?

Suppose there is one **very strong feature** in the data set. When using bagged trees, most of the trees will use that feature as the top split, resulting in an ensemble of similar trees that are **highly correlated**.

Averaging highly correlated quantities does not significantly reduce variance (which is the entire goal of bagging).

By randomly leaving out candidate features from each split, **random forests "decorrelate" the trees** to the extent that the averaging process can reduce the variance of the resulting model.

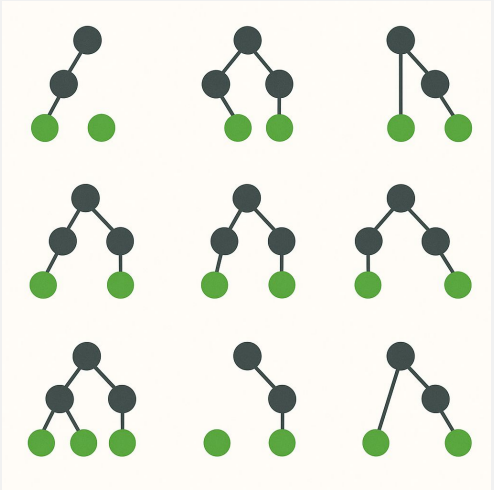
 © 2025 General Assembly

Another way of looking at it...

Sometimes one or two strong features dominate every tree in bagging, resulting in essentially the same tree as every predictor.

(This is what was meant when saying the trees could be highly correlated.)

By using a subset of features to generate each tree, we get a wider variety of predictive trees that do not all use the same dominant features.



 © 2025 General Assembly

Random Forest

The steps for creating a Random Forest model are very similar to Bagging:

1. Get a bootstrap sample
2. Fit a decision tree:
    a. Each split in the tree is based on a **random** subset of m features from all p predictors
    b. Only one of the m features is used for the split
3. Repeating 1 and 2 many times (B = 100 or 500 times)
4. The final model is the average of all B models.

When $m = p$ the Random Forest is a vanilla Bagging model

 © 2025 General Assembly

Random Forest Simulation... Go!

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} & x_{15} & x_{16} \\ x_{21} & x_{22} & x_{23} & x_{24} & x_{25} & x_{26} \\ x_{31} & x_{32} & x_{33} & x_{34} & x_{35} & x_{36} \\ x_{41} & x_{42} & x_{43} & x_{44} & x_{45} & x_{46} \\ x_{51} & x_{52} & x_{53} & x_{54} & x_{55} & x_{56} \\ x_{61} & x_{62} & x_{63} & x_{64} & x_{65} & x_{66} \\ x_{71} & x_{72} & x_{73} & x_{74} & x_{75} & x_{76} \\ x_{81} & x_{82} & x_{83} & x_{84} & x_{85} & x_{86} \\ x_{91} & x_{92} & x_{93} & x_{94} & x_{95} & x_{96} \end{bmatrix}$$

x2

x2

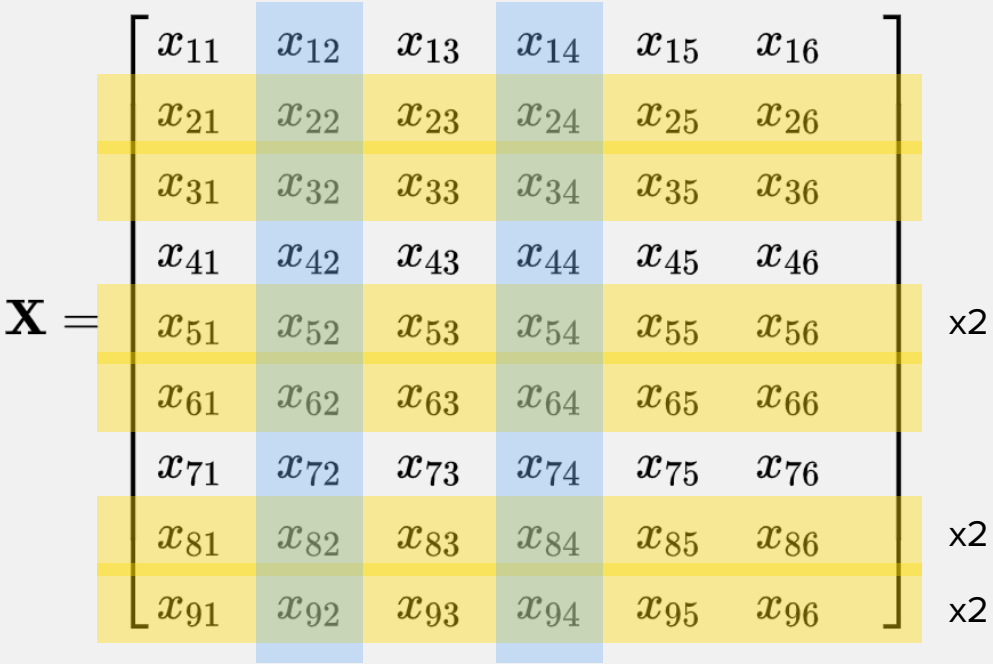
x2

Our bootstrap sample has:

- 9 observations
- 6 features

 © 2025 General Assembly

Random Forest Simulation... Go!

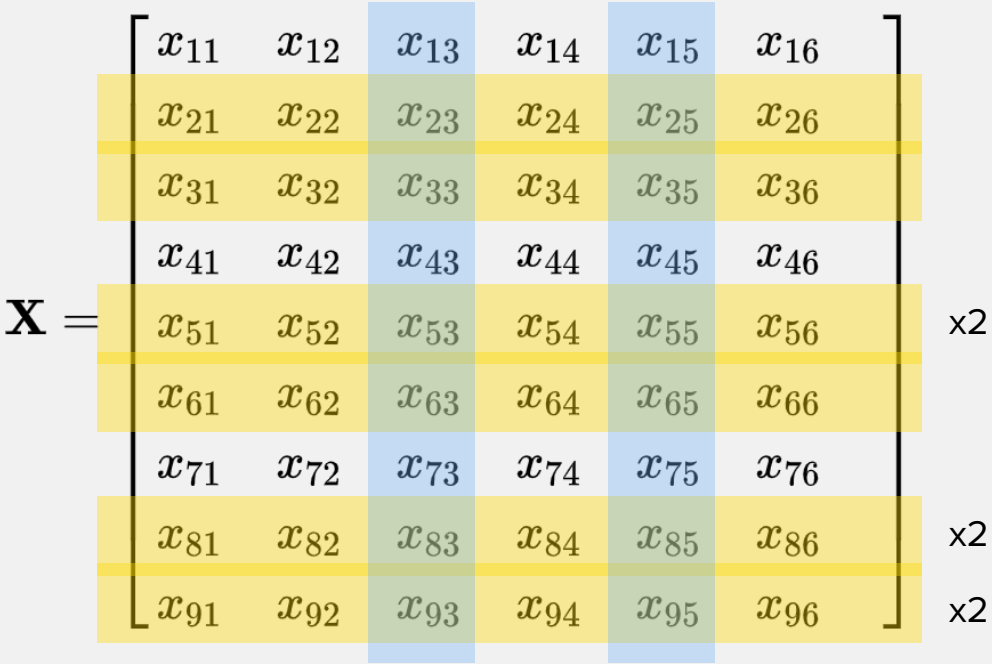


Our bootstrap sample has:

- 9 observations
- 6 features

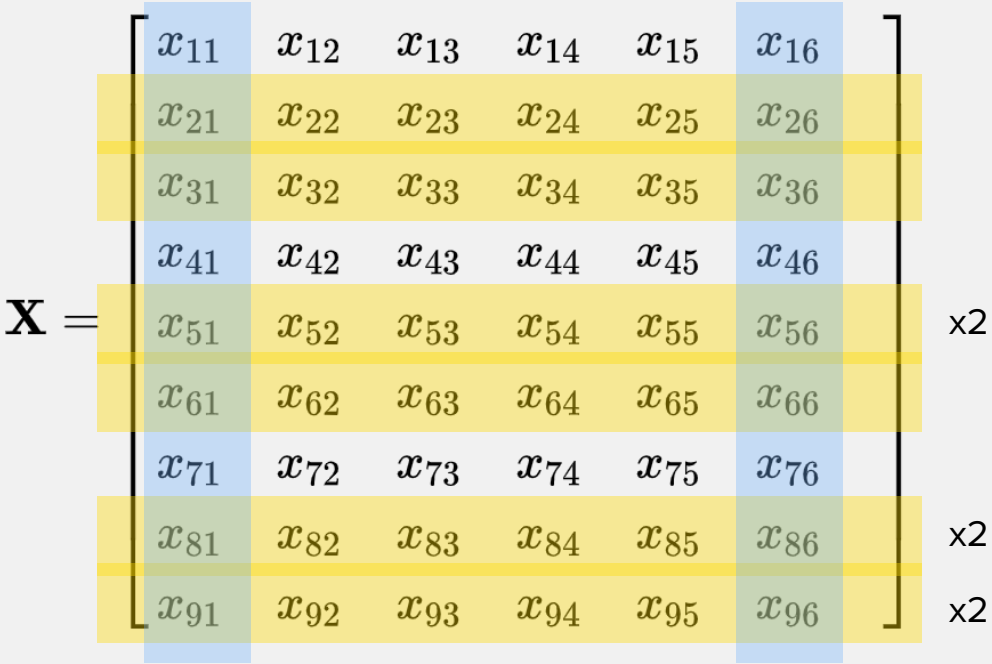
 © 2025 General Assembly

Random Forest Simulation... Go!



 © 2025 General Assembly

Random Forest Simulation... Go!



 © 2025 General Assembly

Random Forest Simulation... Go!

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & x_{13} & x_{14} & x_{15} & x_{16} \\ x_{21} & x_{22} & x_{23} & x_{24} & x_{25} & x_{26} \\ x_{31} & x_{32} & x_{33} & x_{34} & x_{35} & x_{36} \\ x_{41} & x_{42} & x_{43} & x_{44} & x_{45} & x_{46} \\ x_{51} & x_{52} & x_{53} & x_{54} & x_{55} & x_{56} \\ x_{61} & x_{62} & x_{63} & x_{64} & x_{65} & x_{66} \\ x_{71} & x_{72} & x_{73} & x_{74} & x_{75} & x_{76} \\ x_{81} & x_{82} & x_{83} & x_{84} & x_{85} & x_{86} \\ x_{91} & x_{92} & x_{93} & x_{94} & x_{95} & x_{96} \end{bmatrix}$$

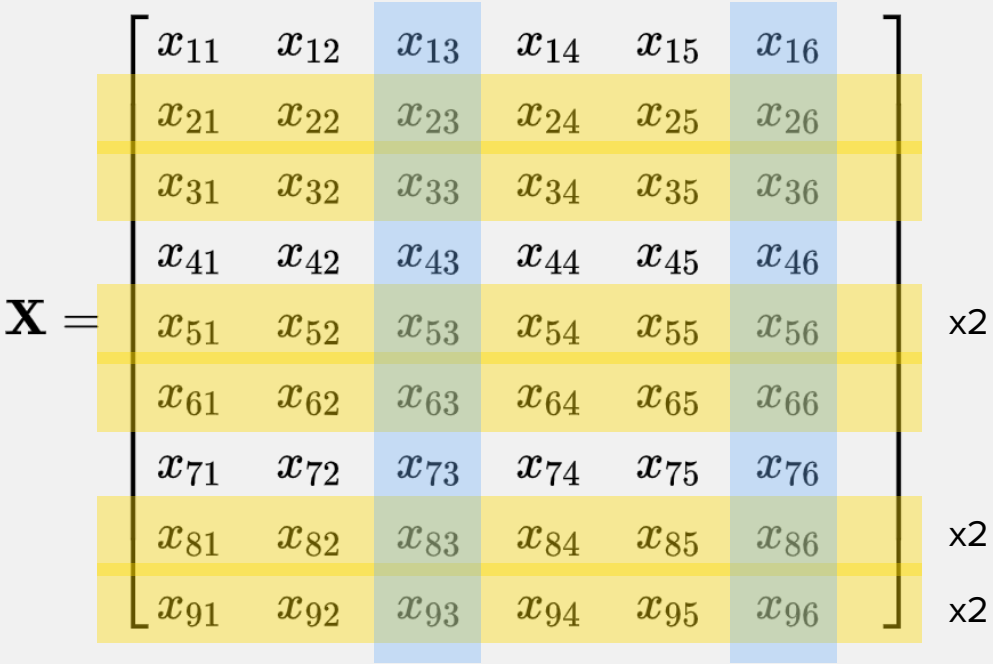
x2

x2

x2

 © 2025 General Assembly

Random Forest Simulation... Go!



 © 2025 General Assembly

Random Forests

Pros:

- Their performance is competitive with the best supervised learning methods.
- They provide a more reliable estimate of feature importance.
- They allow you to estimate out-of-sample error without using train/test split or cross-validation. *(But you still need a TTS split because you're tuning parameters!)*

Cons:

- They are less interpretable.
- They are slower to train.
- They are slower to predict.

 © 2025 General Assembly

Random Forests

Let's see it for ourselves!

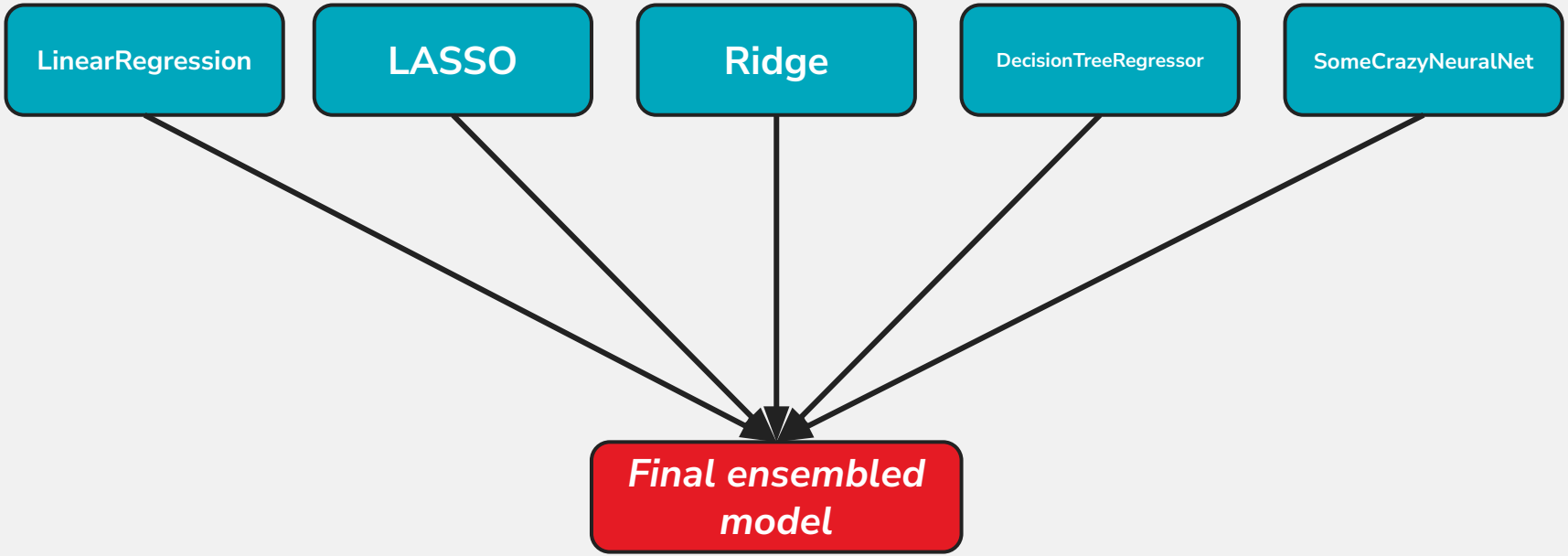
- [ ] Open **02-ensembles-bagging-random-forest.ipynb**



 © 2025 General Assembly

Bring It Home

Stacking involves arbitrary models ensembled together



 © 2025 General Assembly

Key takeaways



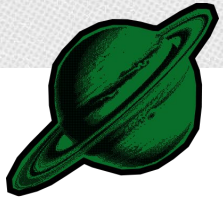
Bagging can improve model performance

It takes an aggregate of bootstrapped weaker models



Random Forest is a type of bagging model

A random subset of features is used for selecting each split (which decorrelates the trees)



Out-of-sample-error can be useful

It uses unsampled data to calculate standard classification metrics

 © 2025 General Assembly


GA
GENERAL ASSEMBLY